'*Memory and Games - Trust and Ultimatum*'

Not public yet

AsPredicted #: 286,187

Pre-registered on
2026/04/20 01:12 (PT)

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1) Have any data been collected for this study already?
No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?
We aim to study whether mnemonic associations causally affect participants' representations of strategic interactions in the Ultimatum Game (UG) and the Trust Game (TG), and whether variation in such representations is correlated with players' behavior in the games and their beliefs about the other player's behavior.

Our main hypotheses are:
1. Manipulating mnemonic associations through (i) reading and rehearsal of a story and (ii) changes in game description, affects representations, as measured by classified free-text justifications of decisions.

2. Representations are correlated with behavior: variation in participants' representations is correlated with their behavior in the games, consistent with the mediating role of representation in control.

3. Part of the treatment effect on behavior operates through representations: differences in mean behavior across conditions can be decomposed into a part due to differences in the distribution of representations and a part due to differences in behavior conditional on each representation category.

3) Describe the key dependent variable(s) specifying how they will be measured.
There are two key classes of dependent variables.
1) Behavioral Outcomes
In the Ultimatum Game:
• Sender: share of the budget offered to the receiver, and belief about the probability that the receiver accepts their offer.
• Receiver: whether they accept the sender's offer.
In the Trust Game:
• Sender: share of the budget sent to receiver, and belief about the amount that the receiver will return to the sender.
• Receiver: amount returned to the sender (over amount received).
For these outcomes, we care about measures of central tendency (mean, median), the distribution of outcomes, and the share of responses at salient values (see Q8).
For both games, we also study the accuracy of senders' beliefs relative to observed receivers' behavior.
2) Game Representation Categories
Participants will be asked an open-ended question about what drove their decision (we call this "Reasons"). These

answers will be classified into pre-specified categories using LLMs, with validation by Research Assistants using the same coding rules.
In the Trust Game:
• Sender: (1) no clear justification; (2) generosity / fairness-based reasoning; (3) joint investment / productive partnership reasoning; (4) strategic conflict / betrayal-avoidance reasoning.
• Receiver: (1) no clear justification; (2) generosity / fairness-based reasoning; (3) joint cooperation / partnership reasoning; (4) strategic self-interest / payoff maximization; (5) reciprocity.
In the Ultimatum Game:
• Sender: (1) no clear justification; (2) fairness / moral sharing; (3) payoff maximization / unconstrained proposer; (4) strategic bargaining / acceptance-as-a-function-of-offer reasoning.
• Receiver: (1) no clear justification; (2) fairness / moral reasoning; (3) payoff maximization; (4) status / respect reasoning.
In addition, responses will be classified into Social Proximity categories using the same procedure as in an earlier pre-registration (#261370).

4) How many and which conditions will participants be assigned to?
We manipulate two dimensions.
Manipulation 1, Story: participants are asked to read and summarize a short text describing an allocation problem from a real-world context; this is meant to prompt memory of their experience outside of the laboratory.
• No Story
• Story with workplace context (Bonus)
• Story with charity context (Aid)
Manipulation 2, Game Frame:
• Control Task (or Abstract Frame): participants are presented with the standard abstract version of the game
• Pricing Task (or Market Frame): participants are presented with a market-framed version of the same strategic interaction. This frame is constructed analogously to the Pricing Task from an earlier pre-registration (#261370).
Story and Game Frame are crossed only partially: the three Story conditions are combined with the Control Task frame, while the Pricing Task frame is run only with No Story. This yields four between-subject conditions (pooling together roles and games):
1. No Story, Control Task
2. Bonus Story, Control Task
3. Aid Story, Control Task
4. No Story, Pricing Task
These four conditions are collected separately for each role (Senders and Receivers) in each game (Ultimatum and Trust Game) for a total of 4×2×2 = 16 between-subject experiments in total.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
We will conduct the following analyses.
(i) Effect of manipulations on behavior and beliefs: using regression analysis, we study the effect of manipulations on the behavioral outcomes described above (see Q3).
In particular, in the Ultimatum Game, we study the effect of manipulations on the offer made by the sender and on the chance the receiver accepts the sender's offer; in the Trust Game, we study the effect of manipulations on the amount sent by the sender and on the amount returned by the receiver. In both games, we study the effect of manipulations on beliefs accuracy.
In the sender's analyses, we will control for beliefs. In the receiver's analysis, we will control for the sender's action. We will also study the full distribution of outcomes and the probability of salient choices (see Q8).
(ii) Effect of manipulations on representations: using regression analysis, we study the effect of manipulations on representation categories, as measured by the classified free-text "Reasons" responses described above. This analysis will be carried out separately by game and player role.
(iii) Correlation between representations and behavior: using regression analysis, we study the relationship between representation categories and behavioral outcomes, separately by game and player role.
(iv) Decomposition of effects
We decompose differences in mean behavior between conditions into two parts:
- one due to differences in the distribution of representations;
- one due to differences in behavior conditional on each representation category.
This is done through a symmetrized (Shapley / Oaxaca-Blinder) decomposition, which averages the two possible

paths (changing distributions first or conditional behavior first) to attribute the total difference fairly between the two components.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

All surveys contain:
- an attention check; and
- comprehension questions.

Participants who fail the attention check or who fail twice to answer all the comprehension questions correctly will be screened out. Screened-out participants will not contribute to the total sample size.

7) How many observations will be collected or what will determine sample size?

No need to justify decision, but be precise about exactly how the number will be determined.

We will collect 600 participants in each of the 16 between-subject experiments described above (4 conditions x 2 roles x 2 games) for a total of 4,800 participants in the role of Senders and up to 4,800 participants in the role of Receivers. This is the reason why the number of Receivers might be lower than 4,800: in the Trust Game, the Receiver decides only if the Sender sends a strictly positive amount; therefore, we will recruit a number of Receivers equal to the number of Senders who choose to send more than 0.

8) Anything else you would like to pre-register?

(e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will extend the analyses to specific behavioral outcomes, focusing on salient decisions (for example, equal split offers in the Ultimatum Game; sending nothing, 50% of the budget, or everything in the Trust Game; returning the amount sent, half of the amount received, or nothing in the Trust Game), as well as on the full distribution of choices.
We will compare the distribution of representations across Ultimatum Game senders, Trust Game senders, and the Dictator Game decision-makers from pre-registration #261370 under the Abstract Frame, and analogously under the Market Frame.
We will analyze the accuracy of senders' beliefs about receivers' behavior in a counterfactual scenario. To this end, we will measure Ultimatum Game senders' beliefs about the probability that the receiver would accept a different offer; whether Ultimatum Game receivers would accept a different offer. Trust Game senders' beliefs about the amount that the receiver would return to the sender if the sender chose a different action; and the amount Trust Game receivers would return to a sender that chose a different action.

Finally, we will collect demographics (such as age, gender, employment status, and experience with Prolific) to study heterogeneity in both behavior and representations across participant groups.